Laboratory Exercise 0

Getting Familiar

Process Control Lab

Simulation software: PC-ControLAB

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Contents

1 Running the Program **2**

2 The Feedback Loop **2**

2.1 Initial Values . . . 3

2.2 Manual Mode: Step Change in OUT . . . 3

2.3 Load Disturbance in Manual Mode . . . 4

2.4 Automatic Mode . . . 4

2.5 Setpoint Change to 350 °F . . . 5

2.6 Load Disturbance in Automatic Mode . . . 7

2.7 Fluctuating Load (AutoLoad) . . . 7

2.8 Undamped Oscillations . . . 8

2.9 Low Gain . . . 9

3 Conclusion **9**

Objective

To become familiar with the simulation software.

1. Running the Program

Q. Run PC-ControLAB, then locate and read `QuickStartGuide.pdf` in the installation folder. Briefly summarize what you learned.

Q. Slow down the simulation speed via View | Scroll Rate (mid-value). Which value did you choose, and why is it more practical?

[Insert screenshot: `figs/main_window.png`]

Figure 1: PC-ControLAB main window.

2. The Feedback Loop

Retrieve the strategy via **Control | Retrieve Strategy, Model and Tuning** and select `Feedbck1`. The simulation starts with an initial controller output (OUT) of 35%.

2.1 Initial Values

Question	Answer
What is the process variable (PV) current value?	
What is its range?	
What is the controller output range?	
What is the setpoint (SP)?	

Table 1: Initial loop values.

2.2 Manual Mode: Step Change in OUT

With the controller in **MAN**, set the controller output to 20%.

Q. Can the process be described as static or dynamic?

Q. At which value does PV settle?

Q. After how long does PV appear to settle at this value?

Q. Calculate the time elapsed from the OUT change until PV is *reasonably close* to its final value (i.e., 98% of the *change* has occurred). Show your calculation.

[Insert screenshot: `figs/manual_step.png`]

Figure 2: PV response to a step in OUT to 20% (MAN mode).

2.3 Load Disturbance in Manual Mode

Change the LOAD by pressing **StepIncr** or **StepDecr** twice in rapid succession.

Q. What do you observe?

[Insert screenshot: `figs/manual_load.png`]

Figure 3: Response to a load change in MAN mode.

2.4 Automatic Mode

Switch the controller to **AUTO**.

Q. Does the controller bring the PV back to the setpoint?

[Insert screenshot: `figs/auto_return.png`]

Figure 4: Controller bringing PV back to SP in AUTO mode.

2.5 Setpoint Change to 350 °F

Increase SP to 350 °F and observe the response.

Q. Does it exhibit more or less time variability compared to the process response with no feedback (MAN)?

Q. Do the oscillations die out? (If so, they are called *damped* oscillations.)

Q. Does the PV reach the new SP?

Q. In how many cycles?

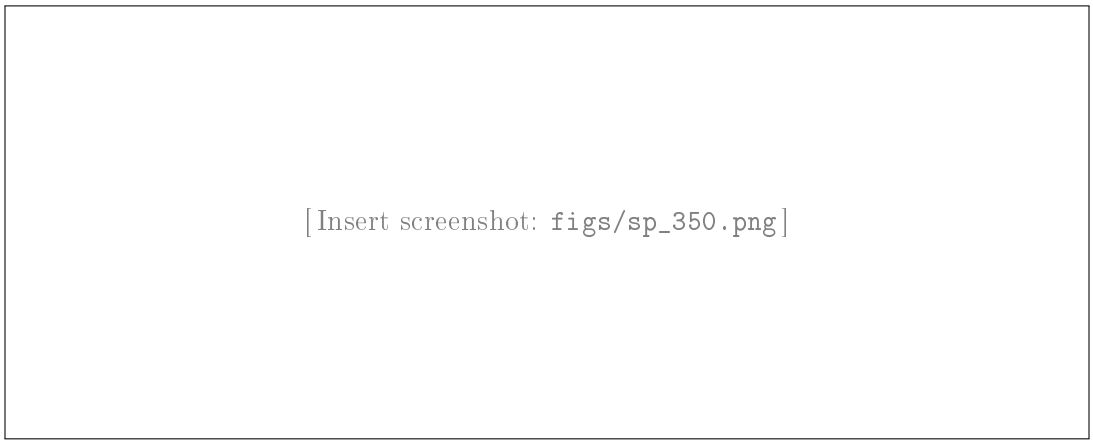


Figure 5: Closed-loop response to SP = 350 °F with response parameters marked.

Parameter	Value
a (peak above final value)	
b (total change, final value − initial value)	
Overshoot $OS = a/b \times 100\%$	
Period of oscillation	

Table 2: Response parameters (see Fig. 5).

Q. What is the PV overshoot? Show your calculation.

Q. What is the period of oscillation?

Q. Does the controller do a good job? ***(For a damped response, a well-tuned controller brings PV to SP in at most one cycle with overshoot not exceeding 5%.)***

2.6 Load Disturbance in Automatic Mode

Change the LOAD by pressing **StepIncr** or **StepDecr** twice in rapid succession.

Q. Does the controller manage to bring the PV back to the setpoint?

[Insert screenshot: `figs/auto_load.png`]

Figure 6: Response to a load change in AUTO mode.

2.7 Fluctuating Load (AutoLoad)

Switch the controller to **MAN** and allow the LOAD to fluctuate (**AutoLoad**). Try to keep PV as close as possible to SP by adjusting OUT for a few minutes.

Q. How easy is this task?

[Insert screenshot: `figs/man_autoload.png`]

Figure 7: Manual control under fluctuating load.

While the LOAD is still fluctuating, switch the controller to **AUTO**.

Q. Observe PV fluctuations around SP. Do they seem acceptable for operation, i.e., confined within a small range (say 1–2% of the PV range)?

[Insert screenshot: `figs/auto_autoload.png`]

Figure 8: Automatic control under fluctuating load.

2.8 Undamped Oscillations

Switch off **AutoLoad**. Keep the controller in AUTO. Click **TUNE** and increase the controller gain until PV exhibits sustained (*undamped*) oscillations.

Quantity	Value
Controller gain for sustained oscillations	

Table 3: Gain at sustained oscillations.

[Insert screenshot: `figs/undamped.png`]

Figure 9: Sustained (undamped) oscillations.

2.9 Low Gain

Reduce the controller gain to 0.1 and change the setpoint again.

Q. What do you observe?

[Insert screenshot: `figs/gain_0p1.png`]

Figure 10: Response with controller gain = 0.1.

3. Conclusion